

Lesson 05: Maximum Likelihood

TTK 4260: Multivariate Data Analysis and ML

Adil RASHEED

23.01.2026



NTNU

Norwegian University of
Science and Technology

Part I: Motivation & Core Concepts

- Why Maximum Likelihood?
- Probability vs Likelihood ← *Key!*
- The Maximum Likelihood Principle

Part II: Distribution Toolkit

- Why Distributions Matter for MLE
- Discrete Distributions
- Continuous Distributions
- Choosing the Right Distribution

Part III: Mathematics of MLE

- Log-Likelihood & Score Function
- MLE for Normal Distribution
- MLE for Other Distributions
- The ML = LS Connection

Part IV: Theory & Applications

- Real-World Applications
- When MLE Fails

Motivation & Core Concepts

Why Maximum Likelihood?

Why Maximum Likelihood?

The fundamental question in data science:

Given observed data, what are the best parameters to describe the underlying process that generated this data?

You already know Least Squares (LS):

$$\theta_{\text{LS}} = \arg \min_{\theta} \sum_{i=1}^n (y_i - f(x_i; \theta))^2$$

Maximum Likelihood (ML) offers something more:

$$\theta_{\text{ML}} = \arg \max_{\theta} P(\text{data} \mid \theta)$$

Key insight: ML incorporates *probabilistic assumptions* about how data is generated.

Why Extend Least Squares to Maximum Likelihood?

What ML adds beyond LS:

- Information about noise distribution
- Uncertainty quantification
- Confidence intervals
- Hypothesis testing framework
- Principled model comparison

The caveat:

- ML “injects” assumptions
- Good assumptions $\Rightarrow$ improved estimates
- Bad assumptions $\Rightarrow$ biased or inefficient estimates

Bottom line: ML is not “better” than LS—it’s *different*. It allows you to leverage domain knowledge about your data’s statistical properties.

When Would You Actually Use This? Real Applications

Quality Control

- Estimate mean & variance of product dimensions
- Compute probability of defects
- Set tolerance limits with confidence

Reliability Engineering

- Estimate component failure rates
- Plan maintenance schedules
- Predict warranty costs

Medical Research

- Estimate treatment effect sizes
- Model survival times
- Clinical trial analysis

Machine Learning

- Train logistic regression
- Fit Gaussian mixtures
- Neural network training (yes, really!)

The Limitations of Least Squares: Why ML Matters

Scenario	LS Problem	ML Solution
Data with outliers	Heavily influenced by outliers	Use heavy-tailed distribution (e.g., Laplacian, Student-t)
Count data (defects, clicks)	Assumes continuous, symmetric errors	Use Poisson distribution
Binary outcomes (pass/fail)	Can predict outside $[0, 1]$	Use Bernoulli/Binomial (gives valid probabilities)
Censored data (time-to-failure)	Ignores censoring mechanism	Use survival distributions (Exponential)

Probability vs Likelihood

The Key Insight: Probability vs Likelihood

Same mathematical function, different perspective:

$$\underbrace{P(\text{data} \mid \text{parameters})}_{\text{probability}} \quad \text{vs} \quad \underbrace{L(\text{parameters} \mid \text{data})}_{\text{likelihood}}$$

Probability:

- Parameters are **fixed** (known)
- Data is **variable** (unknown)
- Question: “Given I know θ , what data might I see?”
- **Before** data collection
- Integrates/sums to 1 over data

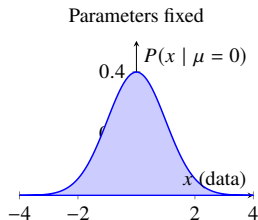
Likelihood:

- Data is **fixed** (observed)
- Parameters are **variable** (unknown)
- Question: “Given this data, which θ is most plausible?”
- **After** data collection
- Does NOT integrate to 1 over θ

Visualizing the Difference

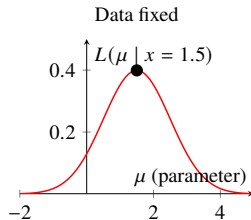
Consider a Normal distribution with unknown mean μ and known $\sigma = 1$

Probability view: Fix $\mu = 0$



Area under curve = 1

Likelihood view: Fix $x = 1.5$



Area under curve $\neq 1$

Maximum likelihood: Find μ that maximizes the red curve $\Rightarrow \hat{\mu}_{ML} = 1.5$

Maximum Likelihood Principle

The Maximum Likelihood Principle

Definition: The Maximum Likelihood Estimator (MLE) is the parameter value that maximizes the likelihood of the observed data.

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta \in \Theta} L(\theta \mid \mathcal{D})$$

In words:

- Among all possible parameter values $\theta \in \Theta$
- Find the one that makes the observed data $\mathcal{D}$ most probable
- This is a **principled** way to choose parameters

Maximum Likelihood vs Least Squares: A Comparison

Least Squares:

$$\hat{\theta}_{\text{LS}} = \arg \min_{\theta} \sum_{i=1}^n (y_i - f(x_i; \theta))^2$$

Interpretation:

- Geometric: minimize distance
- No probabilistic model
- Simple, always applicable
- No uncertainty quantification

Maximum Likelihood:

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta} P(\mathcal{D} \mid \theta)$$

Interpretation:

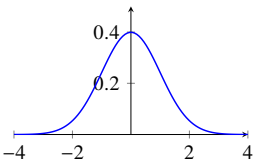
- Statistical: maximize plausibility
- Requires probabilistic model
- More assumptions, more power
- Enables uncertainty quantification

Spoiler: Under Gaussian noise, ML = LS! (We'll prove this soon.)

The Choice of Distribution Matters!

Same data, different distributional assumptions $\Rightarrow$ different estimates

$$L(\mu) \propto e^{-\sum (x_i - \mu)^2 / 2}$$

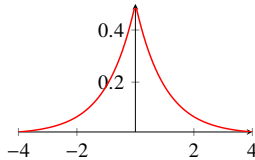


Gaussian assumption

MLE = sample mean $\bar{x}$

Sensitive to outliers

$$L(\mu) \propto e^{-\sum |x_i - \mu|}$$



Laplacian assumption

MLE = sample median

Robust to outliers

Important message: The “best” estimator depends on your assumptions about the data-generating process!

Distribution Toolkit

Why Distributions Matter for MLE

From Distribution to Likelihood Function

The likelihood function comes directly from your distribution choice:

If you assume...	Your likelihood function is...
$X \sim \text{Normal}(\mu, \sigma^2)$	$L(\mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$
$X \sim \text{Poisson}(\lambda)$	$L(\lambda) = \prod_{i=1}^n \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}$
$X \sim \text{Exp}(\lambda)$	$L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i}$
$X \sim \text{Bernoulli}(p)$	$L(p) = \prod_{i=1}^n p^{x_i} (1 - p)^{1-x_i}$

The Takeaway

Different distributions $\Rightarrow$ different likelihood functions $\Rightarrow$ different MLEs!

Bernoulli Distribution

Yes or No? Success or Failure?

Scenario 1: Medical Testing

A rapid test for a disease has 95% accuracy for infected patients.

You test one patient. What's the probability the test is correct?

Scenario 2: Sports

A basketball player has a 80% free throw percentage.

She takes one shot. What's the probability she misses?

Scenario 3: Quality Control

A factory produces items with a 3% defect rate.

You pick one item. How do you model whether it's defective?

What do these questions have in common?

Single trial • Two possible outcomes • Known probability

The Bernoulli Distribution

Setup:

- Single trial with two outcomes
- “Success” with probability p
- “Failure” with probability $1 - p$
- $X \in \{0, 1\}$

Probability Mass Function:

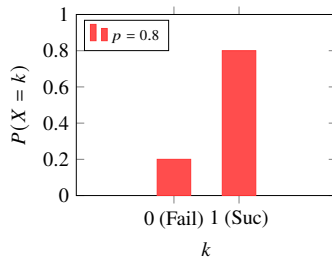
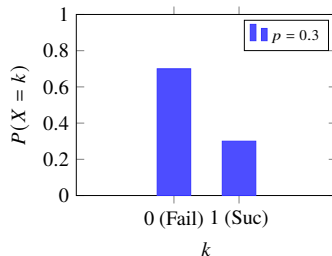
$$P(X = k) = p^k (1 - p)^{1-k}, \quad k \in \{0, 1\}$$

Or simply:

$$P(X = 1) = p$$

$$P(X = 0) = 1 - p$$

PMF for different p



Solving Real Problems

Return to Scenario 2: Free throw percentage is $p = 0.80$.

Probability she misses one shot?

$$P(X = 0) = 1 - p = 1 - 0.80 = \boxed{0.20}$$

Summary Statistics

- Mean: $\mu = p$
- Variance: $\sigma^2 = p(1 - p)$
- Std Dev: $\sigma = \sqrt{p(1 - p)}$

Note: Variance is maximized when $p = 0.5$

The Building Block

- Binomial = sum of n Bernoullis
- Geometric = Bernoullis until first success
- Foundation for binary classification

The Takeaway

The Bernoulli distribution models the simplest random experiment: a single yes/no trial. It is the fundamental building block for more complex distributions.

Binomial Distribution

How Many Successes?

Scenario 1: Quality Control

A factory produces chips with a 5% defect rate. You inspect 100 chips.

What's the probability that at most 3 are defective?

Scenario 2: Clinical Trials

A new drug is effective for 70% of patients. You treat 15 patients.

How likely is it that at least 12 respond positively?

Scenario 3: Sports Analytics

A basketball player has a 40% three-point shooting rate. She takes 8 shots.

What's the probability she makes exactly 5?

What do these questions have in common?

Fixed trials • Two outcomes • Same probability each time

The Binomial Distribution

Setup:

- n independent trials
- Probability p of success each trial
- X = number of successes
- $X \in \{0, 1, 2, \dots, n\}$

Probability Mass Function:

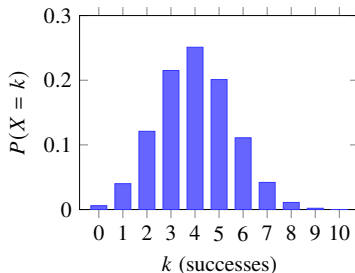
$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$\binom{n}{k}$ ways to arrange k successes

p^k probability of k successes

$(1 - p)^{n-k}$ probability of $n - k$ failures

PMF: $n = 10, p = 0.4$



Mean = $np = 4$

Peak near the mean

Solving Real Problems

Return to Scenario 3: Player shoots 40%, takes $n = 8$ shots.

Probability of exactly $k = 5$ makes?

$$P(X = 5) = \binom{8}{5} (0.4)^5 (0.6)^3 = 56 \times 0.0102 \times 0.216 \approx \boxed{0.124}$$

Summary Statistics

- Mean: $\mu = np$
- Variance: $\sigma^2 = np(1 - p)$
- Std Dev: $\sigma = \sqrt{np(1 - p)}$

Cumulative Probabilities

- $P(X \leq k)$: use tables or software
- $P(X \geq k) = 1 - P(X \leq k - 1)$

The Takeaway

Whenever you count successes across repeated independent trials with constant probability, the binomial distribution gives you exact answers.

Poisson Distribution

How Many Will Occur?

Scenario 1: Traffic Safety

A busy intersection averages 2 accidents per month.

What's the probability of exactly 5 accidents next month?

Scenario 2: Email Management

You receive an average of 12 emails per hour during work.

What's the probability of getting no emails in the next 15 minutes?

Scenario 3: Biology

A biologist counts bacteria in samples. On average, there are 7 per milliliter.

What's the probability a sample contains fewer than 4 bacteria?

What do these questions have in common?

Counting occurrences • Fixed interval • Known average rate

The Poisson Distribution

Setup:

- Events occur independently
- Average rate: λ per interval
- X = count of events in one interval
- $X \in \{0, 1, 2, 3, \dots\}$

Probability Mass Function:

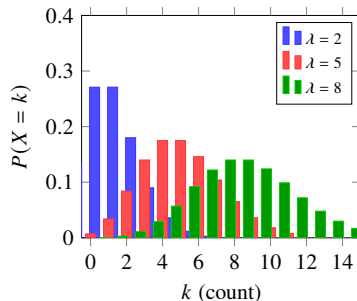
$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

λ^k intensity for k events

$e^{-\lambda}$ probability of “no events”

$k!$ arrangements don't matter

PMF for different λ



Distribution shifts right as λ increases

Key insight: Mean = Variance = λ

Solving Real Problems

Return to Scenario 2: Emails arrive at 12/hour. In 15 min, $\lambda = 12 \times 0.25 = 3$.

Probability of zero emails?

$$P(X = 0) = \frac{3^0 e^{-3}}{0!} = e^{-3} \approx \boxed{0.050}$$

Key Properties for Application:

Summary Statistics

- Mean: $\mu = \lambda$
- Variance: $\sigma^2 = \lambda$
- Std Dev: $\sigma = \sqrt{\lambda}$

Scaling Property

- Rate λ in time t
- Rate $\lambda/2$ in time $t/2$
- Rate 2λ in time $2t$

The Takeaway

Whenever you count *how many* events occur in a fixed window of time or space—with events happening independently at a constant average rate—the Poisson distribution gives you the answer.

Uniform Distribution

When Is Everything Equally Likely?

Scenario 1: A/B Testing

You need to randomly assign each website visitor to group A or group B fairly.

How do you ensure each group has equal probability?

Scenario 2: Simulation

You're running a Monte Carlo stress test with random inputs between 0 and 1.

What's the probability your random number falls between 0.2 and 0.5?

Scenario 3: Random Number Generation

Most programming languages generate random numbers in $[0, 1]$ first.

Why is this interval the starting point for all randomness?

What do these questions have in common?

Equal likelihood • Continuous interval • No value preferred

The Uniform Distribution

Setup:

- Values spread evenly over interval $[a, b]$
- Every sub-interval of equal length has equal probability
- X is continuous, $X \in [a, b]$

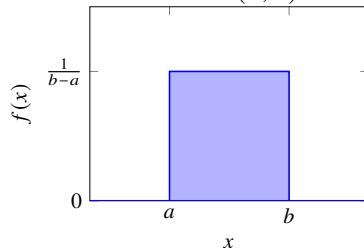
Probability Density Function:

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

Key Probability:

$$P(c \leq X \leq d) = \frac{d - c}{b - a}$$

PDF for Unif(2, 5)



Probability = Area under curve

Total area = 1

Solving Real Problems

Return to Scenario 2: $X \sim \text{Unif}(0, 1)$.

Probability that X falls between 0.2 and 0.5?

$$P(0.2 \leq X \leq 0.5) = \frac{0.5 - 0.2}{1 - 0} = \boxed{0.3}$$

Summary Statistics

- Mean: $\mu = (a + b)/2$
- Variance: $\sigma^2 = \frac{(b - a)^2}{12}$
- Std Dev: $\sigma = \frac{b - a}{\sqrt{12}}$

Why It's Fundamental

- Fair randomization
- Simulation engine
- Transform to other distributions

The Takeaway

The uniform distribution is the foundation of randomness. Computers generate $\text{Unif}(0, 1)$ values, then transform them to create any other distribution.

Exponential Distribution

How Long Until...?

Scenario 1: Customer Service

Customers arrive at a help desk at an average rate of 4 per hour.

What's the probability you wait more than 30 minutes for the next customer?

Scenario 2: Reliability Engineering

A server crashes on average once every 200 hours.

If it just crashed, what's the probability it survives the next 100 hours?

Scenario 3: Call Center

A call center receives calls at a rate of 10 per hour.

What's the expected time until the next call? How variable is this?

What do these questions have in common?

Waiting for an event • Continuous time • Constant average rate

The Exponential Distribution

Setup:

- Events occur at constant rate λ per unit time
- X = waiting time until next event
- X is continuous, $X \geq 0$

Probability Density Function:

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

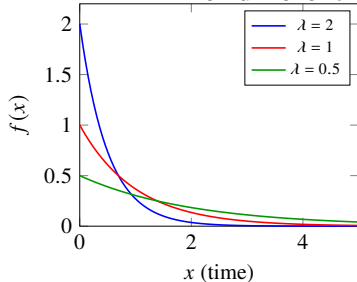
Key Probabilities:

$$P(X > t) = e^{-\lambda t}$$

$$P(X \leq t) = 1 - e^{-\lambda t}$$

Notation: $X \sim \text{Exp}(\lambda)$

PDF for different λ



Higher $\lambda \Rightarrow$

shorter waits **Memoryless Property:**

$$P(X > s + t \mid X > s) = P(X > t)$$

"The system doesn't age"

Solving Real Problems

Return to Scenario 1: Customers arrive at rate $\lambda = 4/\text{hour}$.

Probability of waiting more than 30 min (0.5 hr)?

$$P(X > 0.5) = e^{-\lambda t} = e^{-4 \times 0.5} = e^{-2} \approx \boxed{0.135}$$

Summary Statistics

- Mean: $\mu = \frac{1}{\lambda}$
- Variance: $\sigma^2 = \frac{1}{\lambda^2}$
- Std Dev: $\sigma = \frac{1}{\lambda}$

Useful Formulas

- $P(a < X < b) = e^{-\lambda a} - e^{-\lambda b}$
- Median: $\frac{\ln 2}{\lambda} \approx \frac{0.693}{\lambda}$

The Takeaway

Whenever you measure *time until* an event occurs—and events happen randomly at a constant rate—the exponential distribution models your waiting time.

The Normal (Gaussian) Distribution

Setup:

- μ = mean (center)
- σ = standard deviation (spread)
- X is continuous, $X \in (-\infty, \infty)$

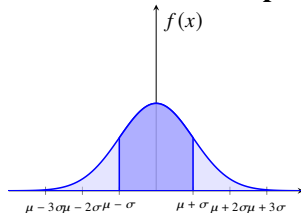
Probability Density Function:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

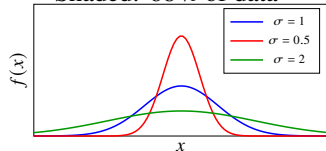
The 68-95-99.7 Rule:

- 68% within $\mu \pm 1\sigma$
- 95% within $\mu \pm 2\sigma$
- 99.7% within $\mu \pm 3\sigma$

The Bell Curve Shape



Shaded: 68% of data



Smaller $\sigma \Rightarrow$ taller, narrower

Solving Real Problems

Return to Scenario 1: Heights have $\mu = 175$ cm, $\sigma = 7$ cm.

What percentage of men are taller than 185 cm?

Step 1: Standardize $Z = \frac{X - \mu}{\sigma} = \frac{185 - 175}{7} = 1.43$

Step 2: Use Z-table $P(X > 185) = P(Z > 1.43) = 1 - 0.9236 \approx \boxed{7.6\%}$

Key Properties for Application:

Summary Statistics

- Mean: μ
- Variance: σ^2
- Median = Mode = μ

Why It's Everywhere

- Central Limit Theorem
- Sum of many small effects
- Natural phenomena

The Takeaway

The normal distribution emerges whenever many independent factors combine additively.

Distribution Choice

Why Distribution Choice Matters for MLE

Key Insight: Maximum Likelihood Estimation (MLE) *assumes* you know the distribution family. Choose wrong, and your estimates can be **biased, inefficient, or meaningless**.

The MLE workflow:

- **Choose a distribution family** ← *This is where insight matters!*
- Write down the likelihood function
- Maximize to find parameter estimates

The problem:

- MLE finds the *best parameters for your chosen distribution*
- It does **not** tell you if you chose the right distribution!
- Garbage in → Garbage out

The Mean-Variance Relationship: A Powerful Tool

Key insight: Each distribution has a characteristic relationship between mean and variance!

Distribution	Variance Formula	What It Means
Normal	σ^2 (constant)	Variance doesn't depend on mean
Poisson	$\sigma^2 = \mu$	Variance equals mean
Binomial	$\sigma^2 = np(1 - p) < np = \mu$	Variance < mean
NegBinom	$\sigma^2 = \mu + \mu^2/r$	Variance > mean
Exponential	$\sigma^2 = \mu^2$	Variance = mean squared

Quick Decision Table

If your data is...	Use...	Check that...
Yes/No outcomes (one trial)	Bernoulli(p)	—
Count of successes in n trials	Binomial(n, p)	Trials are independent
Counts with $s^2 \approx \bar{x}$	Poisson(λ)	Events are independent
Counts with $s^2 > \bar{x}$	NegBinom(r, p)	Overdispersion present
Waiting time (constant hazard)	Exp(λ)	Memoryless property
Positive, right-skewed	Gamma or LogNormal	Check which fits better
Symmetric, unbounded	Normal(μ, σ^2)	Q-Q plot is linear
Symmetric with heavy tails	Student- t	More outliers than Normal
Proportions/probabilities in $[0, 1]$	Beta(α, β)	Bounded on $[0, 1]$
Equally likely in $[a, b]$	Unif(a, b)	No value is preferred

Common Mistakes to Avoid

Mistake 1: “Normal by default”

- Not all data is Normal!
- **Wrong:** Normal for counts, proportions, or times
- **Fix:** Check if data is symmetric and unbounded

Mistake 2: Ignoring bounds

- Probabilities must be in $[0, 1]$
- Times must be ≥ 0
- **Wrong:** Normal for strictly positive data
- **Fix:** Use Exponential, Gamma, or LogNormal

Mistake 3: Not checking assumptions

- Exponential requires memoryless property
- Poisson requires independence
- **Fix:** Test assumptions before choosing

Mistake 4: Discrete vs. Continuous confusion

- Counts $(0, 1, 2, \dots) \neq$ measurements
- **Wrong:** Normal for count data
- **Fix:** Use Poisson, Binomial, etc.

Mistake 5: Ignoring domain knowledge

- Physics/biology often tells you the distribution
- **Fix:** Think about the process first!

How to Verify Your Choice

Visual Methods (start here!)

- **Histogram:** Does the shape match the distribution?
- **Q-Q Plot:** Points should fall on a straight line if the fit is good
- **Mean-Variance Plot:** For count data, plot σ^2 vs. $\bar{x}$

Statistical Tests

- **Shapiro-Wilk:** Test for normality
- **Kolmogorov-Smirnov:** Compare data to any theoretical distribution
- **Chi-squared goodness-of-fit:** For discrete distributions

The Takeaway

Always visualize your data first! Statistical tests can confirm your choice, but your eyes and understanding of the data-generating process should guide you.

Distribution Selection Checklist

Before choosing a distribution, answer these questions:

1 Discrete or continuous?

- Counts $\Rightarrow$ Bernoulli, Binomial, Poisson, Negative Binomial
- Measurements $\Rightarrow$ Normal, Exponential, Gamma, etc.

2 What values are possible?

- $[0, 1] \Rightarrow$ Beta $[a, b] \Rightarrow$ Uniform
- $[0, \infty) \Rightarrow$ Exponential, Gamma, LogNormal
- $(-\infty, \infty) \Rightarrow$ Normal, Student- t

3 What's the mean-variance relationship?

- For counts: $\sigma^2 \approx \bar{x}$? $\sigma^2 > \bar{x}$? $\sigma^2 < \bar{x}$?

4 What process generated the data?

- Sum of effects $\Rightarrow$ Normal
- Product of effects $\Rightarrow$ LogNormal
- Waiting time $\Rightarrow$ Exponential

5 Verify your choice with plots and tests!

Mathematics

Log-Likelihood

Why Use Log-Likelihood?

For **i.i.d. observations** $x_1, x_2, \dots, x_n$, the likelihood is a product:

$$L(\theta) = \prod_{i=1}^n f(x_i \mid \theta)$$

Problem: Products of many small numbers $\rightarrow$ numerical underflow!

Solution: Take the logarithm! Since log is monotonically increasing:

$$\arg \max_{\theta} L(\theta) = \arg \max_{\theta} \log L(\theta) = \arg \max_{\theta} \ell(\theta)$$

The log-likelihood:

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(x_i \mid \theta)$$

Products become sums!

Much easier to work with both analytically and numerically.

Advantages of Log-Likelihood

● Numerical stability:

- Avoids underflow: $\prod_{i=1}^{1000} 0.001 = 10^{-3000}$ (underflows to 0)
- Log version: $\sum_{i=1}^{1000} \log(0.001) = -6908$ (manageable)

● Analytical convenience:

- $\log(ab) = \log a + \log b$ (products $\rightarrow$ sums)
- $\log(e^x) = x$ (exponentials simplify)
- Derivatives of sums easier than derivatives of products

● Optimization:

- Log-likelihood is often concave (bowl-shaped)
- Easier to find global maximum
- Gradient descent works well

● Connection to information theory:

- $-\ell(\theta)/n$ approximates cross-entropy
- ML minimizes KL divergence to true distribution

Finding the MLE: The Score Equation

Strategy: Take derivative, set to zero, solve.

The Score Function:

$$s(\theta) = \frac{\partial}{\partial \theta} \ell(\theta) = \frac{\partial}{\partial \theta} \log L(\theta)$$

The Score Equation (necessary condition):

$$s(\hat{\theta}_{\text{ML}}) = 0$$

Sufficient conditions for $\hat{\theta}$ to be the MLE:

- $\nabla \ell(\hat{\theta}) = \mathbf{0}$ (first-order condition)
- $\nabla^2 \ell(\hat{\theta})$ is negative definite (second-order condition)
- $\hat{\theta}$ is in the interior of Θ
- $\ell(\theta)$ is concave (guarantees global maximum)

Note: These are sufficient, not necessary. Some MLEs exist even when these conditions don't hold.

MLE for Normal Distribution

MLE for Normal Distribution: Setup

Model: $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$

Parameters to estimate: $\theta = (\mu, \sigma^2)$

PDF for single observation:

$$f(x_i \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

Likelihood for all observations (independence $\Rightarrow$ product):

$$L(\mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

This is a product of n terms—let's take the log!

MLE for Normal Distribution: Log-Likelihood Derivation

Step 1: Write the likelihood

$$L(\mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

Step 2: Take the logarithm

$$\begin{aligned}\ell(\mu, \sigma^2) &= \log L(\mu, \sigma^2) \\ &= \sum_{i=1}^n \left[\log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{(x_i - \mu)^2}{2\sigma^2} \right] \\ &= \sum_{i=1}^n \left[-\frac{1}{2} \log(2\pi) - \frac{1}{2} \log(\sigma^2) - \frac{(x_i - \mu)^2}{2\sigma^2} \right]\end{aligned}$$

Step 3: Simplify

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

MLE for Normal Distribution: Finding $\hat{\mu}$

Log-likelihood:

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

Take derivative w.r.t. μ :

$$\frac{\partial \ell}{\partial \mu} = -\frac{1}{2\sigma^2} \cdot 2 \sum_{i=1}^n (x_i - \mu)(-1) = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)$$

Set equal to zero and solve:

$$\begin{aligned} \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) &= 0 \\ \sum_{i=1}^n x_i - n\mu &= 0 \end{aligned}$$

MLE for Normal Distribution: Finding $\hat{\sigma}^2$

Take derivative w.r.t. σ^2 :

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)^2$$

Set equal to zero and solve:

$$\begin{aligned} -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)^2 &= 0 \\ \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)^2 &= \frac{n}{2\sigma^2} \\ \sum_{i=1}^n (x_i - \mu)^2 &= n\sigma^2 \end{aligned}$$

$$\boxed{\hat{\sigma}_{\text{ML}}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

MLE for Normal: Summary and Interpretation

MLEs for Normal Distribution:

$$\hat{\mu}_{\text{ML}} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\hat{\sigma}_{\text{ML}}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

The ML = LS Connection

The Beautiful Connection: ML = LS Under Gaussian Noise

Setup: Linear regression with Gaussian errors

$$y_i = \mathbf{x}_i^\top \boldsymbol{\theta} + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$$

This means: $y_i \mid \mathbf{x}_i \sim \mathcal{N}(\mathbf{x}_i^\top \boldsymbol{\theta}, \sigma^2)$

Log-likelihood:

$$\ell(\boldsymbol{\theta}) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\theta})^2$$

Maximizing $\ell(\boldsymbol{\theta})$ is equivalent to:

$$\arg \max_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta}) = \arg \min_{\boldsymbol{\theta}} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\theta})^2$$

Theorem: Under Gaussian noise, the MLE equals the Least Squares estimator!

$$\hat{\boldsymbol{\theta}}_{\text{ML}} = \hat{\boldsymbol{\theta}}_{\text{LS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Why This Connection Matters

Implications:

- **Justification for LS:** When noise is Gaussian, LS is the “optimal” (ML) estimator
- **When LS fails:** If noise is NOT Gaussian (e.g., heavy-tailed, skewed), LS is no longer optimal
- **Uncertainty quantification:** The ML framework provides:
 - Standard errors for coefficients
 - Confidence intervals
 - Hypothesis tests (t-tests, F-tests)
- **Model comparison:** Likelihood ratio tests, AIC, BIC

LS gives you a point estimate; ML gives you a point estimate PLUS a complete probabilistic framework for inference.

MLE for Other Distributions

MLE for Bernoulli Distribution

Model: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$, where $X_i \in \{0, 1\}$

Likelihood:

$$L(p) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i} = p^{\sum x_i} (1-p)^{n-\sum x_i}$$

Log-likelihood:

$$\ell(p) = \left(\sum_{i=1}^n x_i \right) \log p + \left(n - \sum_{i=1}^n x_i \right) \log(1-p)$$

Solve:

$$\frac{d\ell}{dp} = \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1-p} = 0$$

$$\hat{p}_{\text{ML}} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$$

The sample proportion! (e.g., click-through rate = clicks / impressions)

MLE for Poisson Distribution

Model: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$

Likelihood:

$$L(\lambda) = \prod_{i=1}^n \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} = \frac{\lambda^{\sum x_i} e^{-n\lambda}}{\prod x_i!}$$

Log-likelihood:

$$\ell(\lambda) = \left(\sum_{i=1}^n x_i \right) \log \lambda - n\lambda - \sum_{i=1}^n \log(x_i!)$$

Solve:

$$\frac{d\ell}{d\lambda} = \frac{\sum x_i}{\lambda} - n = 0$$

$$\hat{\lambda}_{\text{ML}} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$$

The sample mean! (e.g., average calls per hour)

MLE for Exponential Distribution

Model: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exponential}(\lambda)$

Likelihood:

$$L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum x_i}$$

Log-likelihood:

$$\ell(\lambda) = n \log \lambda - \lambda \sum_{i=1}^n x_i$$

Solve:

$$\frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^n x_i = 0$$

$$\hat{\lambda}_{\text{ML}} = \frac{n}{\sum_{i=1}^n x_i} = \frac{1}{\bar{x}}$$

The reciprocal of the sample mean! (e.g., failure rate = 1 / mean time to failure)

Pattern Recognition: MLEs Are Often Simple Functions

Distribution	Parameter(s)	MLE
Normal	μ σ^2	$\bar{x}$ $\frac{1}{n} \sum (x_i - \bar{x})^2$
Bernoulli	p	$\bar{x}$ (sample proportion)
Poisson	λ	$\bar{x}$
Exponential	λ	$1/\bar{x}$
Uniform $[0, \theta]$	θ	$\max(x_1, \dots, x_n)$

Why are MLEs often simple?

- These distributions belong to the **exponential family**
- For exponential families, MLEs are functions of **sufficient statistics**
- Sufficient statistics “capture all information” about the parameter

Part IV: Applications

Application 1: Quality Control

Scenario: A factory produces bolts with target diameter 10mm.

Data: Measure 50 bolts: $\bar{x} = 10.02\text{mm}$, $s = 0.15\text{mm}$

Questions ML can answer:

- **Point estimates:**

- $\hat{\mu}_{\text{ML}} = 10.02\text{mm}$ (mean diameter)
- $\hat{\sigma}_{\text{ML}} = 0.15\text{mm}$ (process variability)

- **Confidence interval for mean:**

$$\mu \in 10.02 \pm 1.96 \cdot \frac{0.15}{\sqrt{50}} = [9.98, 10.06] \text{ mm (95\% CI)}$$

- **Probability of defect** (spec: 9.7 – 10.3mm):

$$P(\text{defect}) = P(X < 9.7) + P(X > 10.3) \approx 4.6\%$$

Application 2: A/B Testing (Conversion Rates)

Scenario: Test two website designs for conversion rate.

Data:

- Design A: 1000 visitors, 45 conversions
- Design B: 1000 visitors, 62 conversions

MLEs: $\hat{p}_A = 0.045$, $\hat{p}_B = 0.062$

Is B really better, or just random variation?

Using ML framework + asymptotic normality:

$$\text{SE}(\hat{p}_A - \hat{p}_B) = \sqrt{\frac{\hat{p}_A(1 - \hat{p}_A)}{n_A} + \frac{\hat{p}_B(1 - \hat{p}_B)}{n_B}} \approx 0.0095$$

$$z = \frac{0.062 - 0.045}{0.0095} \approx 1.79 \quad (p\text{-value} \approx 0.07)$$

Conclusion: Suggestive but not statistically significant at 5% level.

Application 3: Predictive Maintenance

Scenario: Estimate machine component failure rate for maintenance scheduling.

Data: 10 components failed at times (hours): 120, 85, 200, 150, 95, 180, 110, 160, 140, 130

Model: Exponential distribution (constant failure rate)

MLE:

$$\hat{\lambda}_{\text{ML}} = \frac{n}{\sum t_i} = \frac{10}{1370} \approx 0.0073 \text{ failures/hour}$$

Mean time to failure: $\text{MTTF} = 1/\hat{\lambda} = 137 \text{ hours}$

Business decisions:

- Schedule preventive maintenance at 100 hours (before most failures)
- $P(\text{failure before 100h}) = 1 - e^{-0.0073 \times 100} \approx 52\%$
- Warranty cost estimation: if warranty is 50 hours, $P(\text{claim}) \approx 31\%$

ML in Machine Learning

Maximum Likelihood is Everywhere in Machine Learning

Surprise: Most of the models you use are trained via Maximum Likelihood!

Model	ML Connection
Linear Regression	ML under Gaussian noise = Least Squares
Logistic Regression	ML for Bernoulli (binary cross-entropy loss)
Softmax Regression	ML for Categorical (cross-entropy loss)
Gaussian Mixture Models	ML via EM algorithm
Hidden Markov Models	ML via Forward-Backward algorithm
Neural Networks	ML with various likelihoods (MSE, cross-entropy)
Naive Bayes	ML estimation of class conditionals

Cross-Entropy Loss = Negative Log-Likelihood

Binary classification: $y \in \{0, 1\}$, model predicts $\hat{p} = P(y = 1|x)$

Bernoulli likelihood:

$$L = \prod_{i=1}^n \hat{p}_i^{y_i} (1 - \hat{p}_i)^{1-y_i}$$

Negative log-likelihood:

$$-\ell = - \sum_{i=1}^n [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)]$$

This is exactly the **binary cross-entropy loss**!

Training a neural network with cross-entropy = Maximum Likelihood estimation!

Implication: All the theory (consistency, asymptotic normality, efficiency) applies to deep

MLE and Information Theory: A Deep Connection

Kullback-Leibler Divergence: measures “distance” between distributions

$$D_{KL}(P\|Q) = \mathbb{E}_P \left[\log \frac{P(X)}{Q(X)} \right]$$

Remarkable fact: MLE minimizes KL divergence!

If P_0 is the true distribution and P_θ is our model:

$$\hat{\theta}_{\text{ML}} = \arg \min_{\theta} D_{KL}(P_0\|P_\theta)$$

Intuition: MLE finds the model distribution that is “closest” to the true data-generating distribution in an information-theoretic sense.

Connection to cross-entropy:

$$\text{Cross-Entropy}(P_0, P_\theta) = H(P_0) + D_{KL}(P_0\|P_\theta)$$

 **MIT** Minimizing cross-entropy = minimizing KL divergence (since $H(P_0)$ is constant).

When ML Fails

When Maximum Likelihood Fails

ML is not a silver bullet. It can fail when:

- **Model misspecification:**

- If true distribution $\neq$ assumed distribution, MLE may be biased
- Example: Using Normal when data is heavy-tailed

- **Small sample size:**

- Asymptotic properties don't hold
- MLE may be biased (e.g., $\hat{\sigma}^2$ for Normal)

- **Boundary issues:**

- MLE at boundary of parameter space
- Example: Uniform $[0, \theta]$, MLE = $\max(x_i)$ is at boundary

- **Non-identifiability:**

- Multiple parameters give same likelihood
- Example: Mixture models with label switching

- **Computational issues:**

- Multiple local maxima
- No closed-form solution

Example: MLE Failure with Heavy-Tailed Data

Scenario: Data with occasional outliers (heavy-tailed)

True model: t -distribution with 3 degrees of freedom

Assumed model: Normal distribution

Problem: Normal MLE ($\bar{x}$) is heavily influenced by outliers

Data: 1, 2, 2, 3, 3, 3, 4, 4, 5, 50

Normal MLE: $\hat{\mu} = 7.7$

The outlier dominates!

Better approaches:

- Use t -distribution (heavier tails)
- Use Laplacian (MLE = median)
- Robust estimation methods

Lesson: The “best” estimator depends on your assumptions. Wrong assumptions can lead to poor estimates!

Sampling Strategies

Sampling Strategies: Getting Good Data for ML

ML estimation is only as good as your data!

Simple Random Sampling

- Each member has equal probability
- Gold standard for unbiased estimation
- May be impractical for large populations

Systematic Sampling

- Select every k -th item
- Easy to implement
- Can introduce bias if periodic patterns exist

Stratified Sampling

- Divide into subgroups (strata)
- Sample proportionally from each
- Reduces variance, ensures representation

Cluster Sampling

- Divide into clusters, sample entire clusters
- Practical for geographically dispersed populations
- May increase variance

Summary

Key Takeaways

- **ML Principle:** Choose parameters that make observed data most probable

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta} P(\text{data} \mid \theta)$$

- **Log-likelihood:** Always work with $\ell(\theta) = \log L(\theta)$ for numerical stability
- **ML = LS:** Under Gaussian noise, Maximum Likelihood equals Least Squares
- **Properties:** MLEs are consistent, asymptotically normal, and efficient
- **Fisher Information:** Quantifies precision; enables confidence intervals
- **Invariance:** MLE of $g(\theta)$ is $g(\text{MLE of } \theta)$
- **Applications:** Quality control, A/B testing, predictive maintenance, machine learning
- **Caution:** ML requires correct distributional assumptions!

Thank you!

Questions?

Adil Rasheed
adil.rasheed@ntnu.no